Link with z-transform

Recall that the z-transform of a discrete-signal $x[n]$ is defined as

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n} \longrightarrow (1)$$

And the Fourier transform equation is given by

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \longrightarrow (2)$$

Comparing equation (2) & (1), it is clear that Fourier transform is a special case of z-transform with $z = e^{j\omega}$, i.e. the z-transform evaluated on the unit circle.

This is similar to the relation between Laplace transform and CT-Fourier transform.

Hence, the properties of DTFT directly follows from the properties of z-transform. We donot prove these properties again as we have already done this for z-transform. (Check page 4 module-3)

Properties of DTFT:

Let $x_1[n] \xleftrightarrow{\mathcal{F}} X_1(e^{j\omega})$

$x_2[n] \xleftrightarrow{\mathcal{F}} X_2(e^{j\omega})$

① Linearity:

$$a x_1[n] + b x_2[n] \xleftrightarrow{\mathcal{F}} a X_1(e^{j\omega}) + b X_2(e^{j\omega})$$

② Time shift:

$$x_1[n-n_0] \xleftrightarrow{\mathcal{F}} e^{-j\omega n_0} X_1(e^{j\omega})$$

③ Conjugation & Conjugate Symmetry:

$$x_1^*[n] \xleftrightarrow{\mathcal{F}} X^*(e^{-j\omega})$$

④ Differencing and accumulation:

$$x_1[n] - x_1[n-1] \xleftrightarrow{\mathcal{F}} X(e^{j\omega})(1-e^{-j\omega})$$

Try proving this $\leftarrow$ $$\sum_{m=-\infty}^{\infty} x[m] \xleftrightarrow{\mathcal{F}} \frac{1}{1-e^{-j\omega}} X(e^{j\omega}) + \pi X(e^{j0}) \sum_{k=-\infty}^{\infty} \delta(\omega - 2k)$$

⑤ Time reversal:

$$x_1[-n] \xleftrightarrow{\mathcal{F}} X(e^{-j\omega})$$

⑥ Differentiation in frequency:

$$n x[n] \xleftrightarrow{F} j \frac{d}{d\omega} X(e^{j\omega})$$

⑦ Parseval's relation:

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{2\pi} |X(e^{j\omega})|^2 d\omega$$

Parseval's relation states that ~~this~~ ~~ene~~ the total energy in a signal $x[n]$ (given by the left hand side of above equation) can also be determined by integrating the energy per unit frequency, $|X(e^{j\omega})|^2/2\pi$, over a full 2π interval of distinct discrete-time frequencies

⑧ Convolution:

$$x_1[n] * x_2[n] \xleftrightarrow{F} X_1(e^{j\omega}) X_2(e^{j\omega})$$

⑨ Multiplication property:

$$x_1[n]\, x_2[n] \xleftrightarrow{F} \frac{1}{2\pi} \int_{2\pi} X_1(e^{j\theta}) X_2(e^{j(\omega-\theta)}) d\theta$$

Proof: Let $Y(e^{j\omega})$ be the DTFT of $x_1[n]x_2[n]$.

$$\therefore \; Y(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x_1[n]\, x_2[n]\, e^{-j\omega n}$$

$$\because \; x_1[n] = \frac{1}{2\pi}\int_{2\pi} X_1(e^{j\theta})\, e^{j\theta n}\, d\theta$$

$$\therefore \; Y(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x_2[n] \left\{ \frac{1}{2\pi}\int_{2\pi} X_1(e^{j\theta})\, e^{j\theta n}\, d\theta \right\} e^{-j\omega n}$$

Interchanging sum & integration, we obtain

$$Y(e^{j\omega}) = \frac{1}{2\pi}\int_{2\pi} X_1(e^{j\theta}) \left[\sum_{n=-\infty}^{\infty} x_2[n]\, e^{-j\omega n} \cdot e^{j\theta n} \right] d\theta$$

$$= \frac{1}{2\pi}\int_{2\pi} X_1(e^{j\theta}) \left[\sum_{n=-\infty}^{\infty} x_2[n]\, e^{-j(\omega-\theta)n} \right] d\theta$$

$$\therefore \; \boxed{Y(e^{j\omega}) = \frac{1}{2\pi}\int_{2\pi} X_1(e^{j\theta})\, X_2\left(e^{j(\omega-\theta)}\right) d\theta}$$

(10) Duality:

Recall from Module 8b page 2, that we talked about duality between the fourier transform and inverse fourier transform because of the similarity of the two expressions. But in case of discrete-time fourier transform and its inverse there is no similarity of the expressions — one of them is a sum and the other an integral. So, what kind of duality are we taking about here. Let us revisit all the formulas that we have studied till now —

Fourier →	PERIODIC (Series)	APERIODIC (Transform)
DISCRETE	$x[n] = \sum_{k=\langle N\rangle} a_k e^{j\omega_0 k n} \rightarrow$ (A) $a_k = \frac{1}{N}\sum_{n=\langle N\rangle} x[n] e^{-jk\omega_0 n} \rightarrow$ (B) $\omega_0 = 2\pi/N$	$x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega \rightarrow$ (i) $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \rightarrow$ (ii)
CONTINUOUS	$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{j\omega_0 k t} \rightarrow$ (C) $a_k = \frac{1}{T}\int_T x(t) e^{-j\omega_0 k t} dt \rightarrow$ (D) $\omega_0 = 2\pi/T$	$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} dt \rightarrow$ (iii) $X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} d\omega \rightarrow$ (iv)

Notice that equation (i) is similar to equation (D) and equation (ii) is similar to equation (C).

Recall that $X(e^{j\omega})$ is a periodic function of ω with period 2π. So, we can find the Fourier series representation of $X(e^{j\omega})$. It's Fourier series representation is a weighted sum of harmonically related periodic exponential functions of ω, all of which have the common period of 2π. That is, $X(e^{j\omega})$ can be represented in a Fourier series as a weighted sum of the signals $e^{j\omega n}$, $n = 0, \pm 1, \pm 2, \ldots$ From equation (i) we can rewrite it as —

$$x[-n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega})\, e^{-j\omega(-n)}\, d\omega \longrightarrow (3)$$

Note that $\because$ period of $X(e^{j\omega})$ is 2π. $\therefore \omega_0 = \frac{2\pi}{T} = \frac{2\pi}{2\pi} = 1$

$\therefore$ (3) $\Rightarrow$

$$x[-n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega})\, e^{-j\omega_0(-n)\omega}\, d\omega$$

Replacing $-n = k$,

$$x[k] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega})\, e^{-j\omega_0 k\omega}\, d\omega$$

Notice this is exactly same as equation (D) with $T = 1$ and $\omega \to t$.

What does this analysis tell us.

If we take a signal $x[n]$ and find it's Fourier transform we get $X(e^{j\omega})$. $X(e^{j\omega})$ is a periodic signal with period 2π. The conitinuous time Fourier series (CTFS) coefficients of $X(e^{j\omega})$ is just the time reversal of $x[n]$, i.e., $x[-n]$.

<u>Example</u>: Recall from module-9a page 7, the Fourier transform (DTFT) of

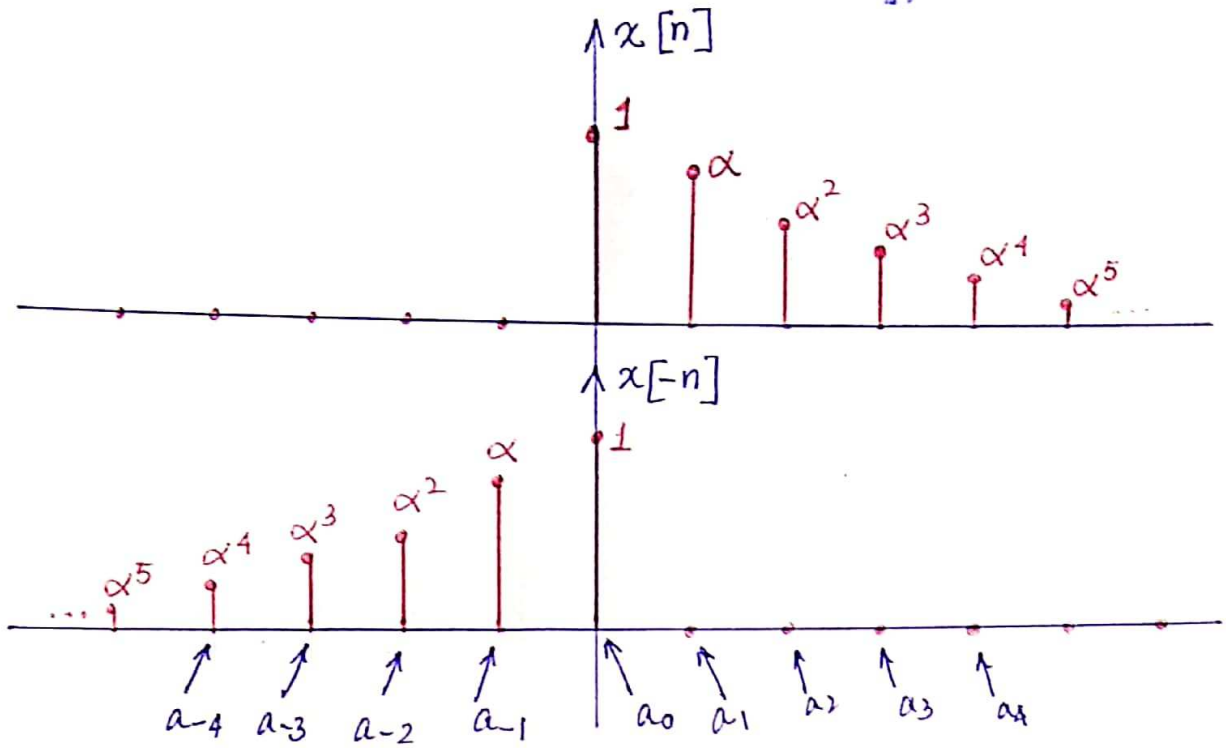
$$x[n] = \alpha^n u[n] \quad , \quad |\alpha| < 1$$

is

$$X(e^{j\omega}) = \frac{1}{1 - \alpha e^{-j\omega}}$$

According to our analysis, the Fourier series co-efficients of $X(e^{j\omega})$ are —

$$x[-n] = \alpha^{-n} u[-n], \text{ i.e.}$$



$\therefore$ The Fourier series co-efficients of $X(e^{j\omega})$ are

$$a_k = \alpha^{-k} u[-k], \quad |\alpha| < 1.$$

The duality that we have talked about till now are —

1. (iii) $\Leftarrow$ dual $\Rightarrow$ (iv)
2. (i) $\Leftarrow$ dual $\Rightarrow$ (D)
3. (A) $\Leftarrow$ dual $\Rightarrow$ (B) $\leftarrow$ We have not discussed this in DTFS but I think you all can figure this one out.